

How did I come up with this code?:

For one of our lab assignments, we had to write down how we were going to use a for loop and the logical XOR operator. Through this pseudocode breakdown, I was able to find resources to try and translate this pseudo code into assembly language to perform the comparison between Foo and Bar.

This was my pseudocode breakdown:

Have two strings loaded in, use a for loop to break down the strings into individual characters. After getting the first character of both strings, XOR the two characters to see if they are the same or different and count the differing bits, then move to the next character and do the same process until we run out of characters. Once we are out of characters, exit the loop and print the result.

Once I had the pseudocode, I translated the steps into assembly and commented on my work so I could understand what was happening where. Also, by using a FOR loop to break down the characters, I should be able to do comparisons with large strings.

Code:

```
section .data
    str1 db "Foo", 0    ; First string
    str2 db "Bar", 0    ; Second string
    msg  db "The hamming distance is: ", 0
    len_msg equ $ - msg
    newline db 10, 0

section .bss
    result resb 4
    count  resb 4

section .text
    global _start

_start:
    xor ecx, ecx        ; ECX will be our loop counter (index)
    xor eax, eax        ; Clear EAX for result count

loop_chars:
    mov dl, [str1 + ecx] ; Load character from str1
    mov dh, [str2 + ecx] ; Load character from str2
    test dl, dl          ; Check if we hit null terminator
```

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    jz print_result      ; If so, exit loop
    xor dl, dh           ; XOR both characters
    call count_bits      ; Count differing bits
    inc ecx              ; Move to next character
    jmp loop_chars       ; Repeat

count_bits:
    push ecx             ; Save ECX
    mov ecx, 8
    xor ebx, ebx         ; Clear bit count

bit_count_loop:
    test dl, 1           ; Check least significant bit
    jz skip_inc          ; If 0, skip increment
    inc ebx              ; Otherwise, increase count

skip_inc:
    shr dl, 1            ; Shift right to check next bit
    loop bit_count_loop  ; Loop for all 8 bits

    add eax, ebx         ; Accumulate total differences
    pop ecx              ; Restore ECX
    ret

print_result:
    ; Convert number to string
    mov edi, result
    mov ebx, eax
    add ebx, '0'         ; Convert to ASCII
    mov [edi], ebx
    mov byte [edi+1], 0 ; Null terminate

    ; print the base string message
    mov eax, 4
    mov ebx, 1
    mov ecx, msg
    mov edx, len_msg
    int 0x80

    ; print the result from hamming
    mov eax, 4
    mov ebx, 1
    mov ecx, result

```

```

mov edx, 1
int 0x80

; print new line
mov eax, 4
mov ebx, 1
mov ecx, newline
mov edx, 1
int 0x80

mov eax, 1
xor ebx, ebx
int 0x80

```

Result with “Foo” and “Bar”:

```

1 section .data
2     str1 db "Foo", 0    ; First string
3     str2 db "Bar", 0    ; Second string
4     msg  db "The hamming distance is: ", 0
5     len_msg equ $ - msg
6     newline db 10, 0
7
8 section .bss
9     result resb 4

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENT

```

Connecting to UMBCgl...
[spourif1@linux4 hemmingdistancelab]$ ./hamming
The hamming distance is: 8
[spourif1@linux4 hemmingdistancelab]$

```

Result with “Foo” and “Foo”

```
1  section .data
2      str1 db "Bang", 0 ; First string
3      str2 db "Fang", 0 ; Second string
4      msg db "The hamming distance is: ", 0
5      len_msg equ $ - msg
6      newline db 10, 0
7
8  section .bss
9      result resb 4
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS

Connecting to UMBCgl...
[spourif1@linux4 hemmingdistancelab]\$./hamming
The hamming distance is: 8
[spourif1@linux4 hemmingdistancelab]\$ nasm -f elf32 -g hamming.asm
[spourif1@linux4 hemmingdistancelab]\$ ld -m elf_i386 -o hamming hamming.o
[spourif1@linux4 hemmingdistancelab]\$./hamming
The hamming distance is: 0
[spourif1@linux4 hemmingdistancelab]\$ nasm -f elf32 -g hamming.asm
[spourif1@linux4 hemmingdistancelab]\$ ld -m elf_i386 -o hamming hamming.o
[spourif1@linux4 hemmingdistancelab]\$./hamming
The hamming distance is: 1
[spourif1@linux4 hemmingdistancelab]\$

Result between “Fang” and “Bang”

```
1  section .data
2      str1 db "Bang", 0 ; First string
3      str2 db "Fang", 0 ; Second string
4      msg db "The hamming distance is: ", 0
5      len_msg equ $ - msg
6      newline db 10, 0
7
8  section .bss
9      result resb 4
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS COMMENTS

Connecting to UMBCgl...
[spourif1@linux4 hemmingdistancelab]\$./hamming
The hamming distance is: 8
[spourif1@linux4 hemmingdistancelab]\$ nasm -f elf32 -g hamming.asm
[spourif1@linux4 hemmingdistancelab]\$ ld -m elf_i386 -o hamming hamming.o
[spourif1@linux4 hemmingdistancelab]\$./hamming
The hamming distance is: 0
[spourif1@linux4 hemmingdistancelab]\$ nasm -f elf32 -g hamming.asm
[spourif1@linux4 hemmingdistancelab]\$ ld -m elf_i386 -o hamming hamming.o
[spourif1@linux4 hemmingdistancelab]\$./hamming
The hamming distance is: 1
[spourif1@linux4 hemmingdistancelab]\$